



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Even Semester)

Degree, Semester & Branch: B.E.VI Computer Science and Engineering

Course Code & Title: CCS361 – Robotic Process Automation

Name of the Faculty member (s): Dr.M. Swarna Sudha

Innovative Practice Description

Unit III – Topic: Scraping

Course Outcome: CO3

Topic Learning Outcome: 3.2

Activity Chosen: Experiential Learning – Real-Time Web Data Scraping Activity

Justification:

Experiential learning provides an opportunity to learn concepts through direct hands-on experience. The scraping activity enables learners to understand the process of extracting structured data from websites and applications. The activity strengthens analytical thinking, practical implementation skills, and problem-solving abilities through real-time interaction with automation tools.

Time Allotted for the Activity: 30 Minutes

Details of the Implementation:

- Participants were divided into five groups for collaborative learning and practical implementation activities.
- Different web-based scenarios were assigned to each group such as:
 - Online product details extraction
 - News headline collection
 - Movie information extraction
 - Book details extraction
 - Job posting information extraction
- Each group identified:
 - Target website information
 - Data elements to be extracted
 - Input and output requirements
 - Scraping workflow sequence
 - Data validation process
- One participant from each group demonstrated the scraping process while other members handled workflow creation, data collection, and result verification.
- Groups designed workflow steps for scraping activities using automation concepts.

- Participants extracted sample information and analyzed the collected data.
- Faculty consolidated the concepts and discussed the importance of data scraping and workflow optimization

CO – PO / PSO mapping:

CO	PO4	PSO2
CO2	2	2

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO4	PSO2
	2	2
Justification for correlation	Ability to analyze requirements and design suitable solutions for data extraction processes.	Ability to develop practical automation-based solutions for real-world applications.

Screenshot of the practice:

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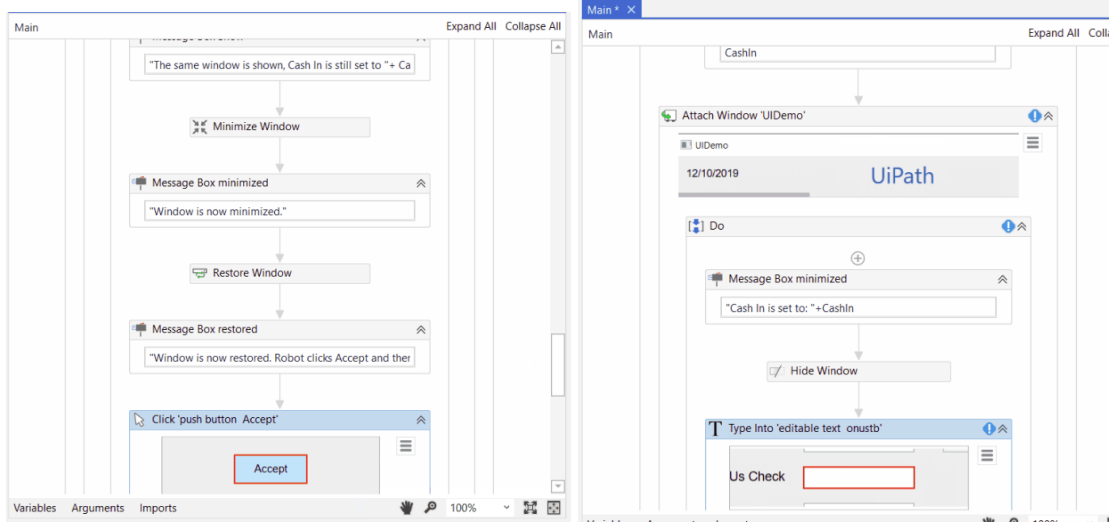


Figure 1 Implementation of Scraping Workflow using UiPath Activities

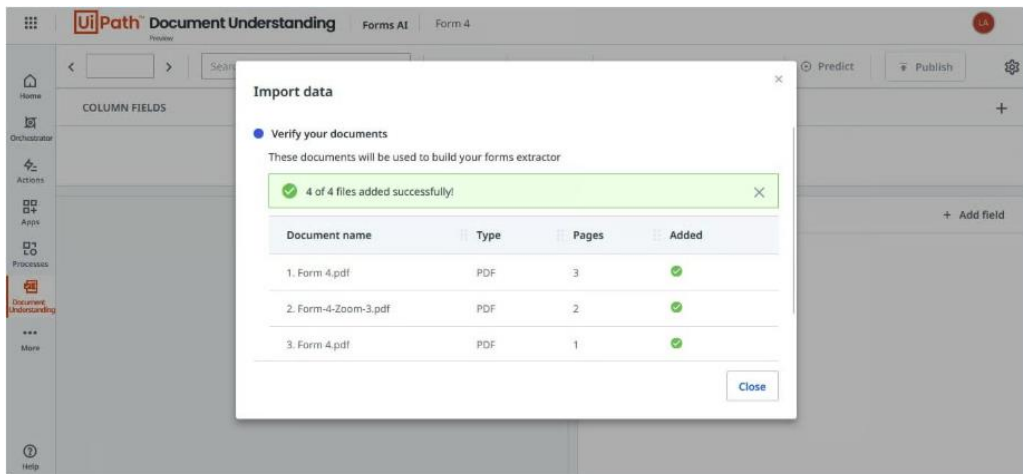


Figure 2 Verification of Input Documents for Data Extraction using UiPath

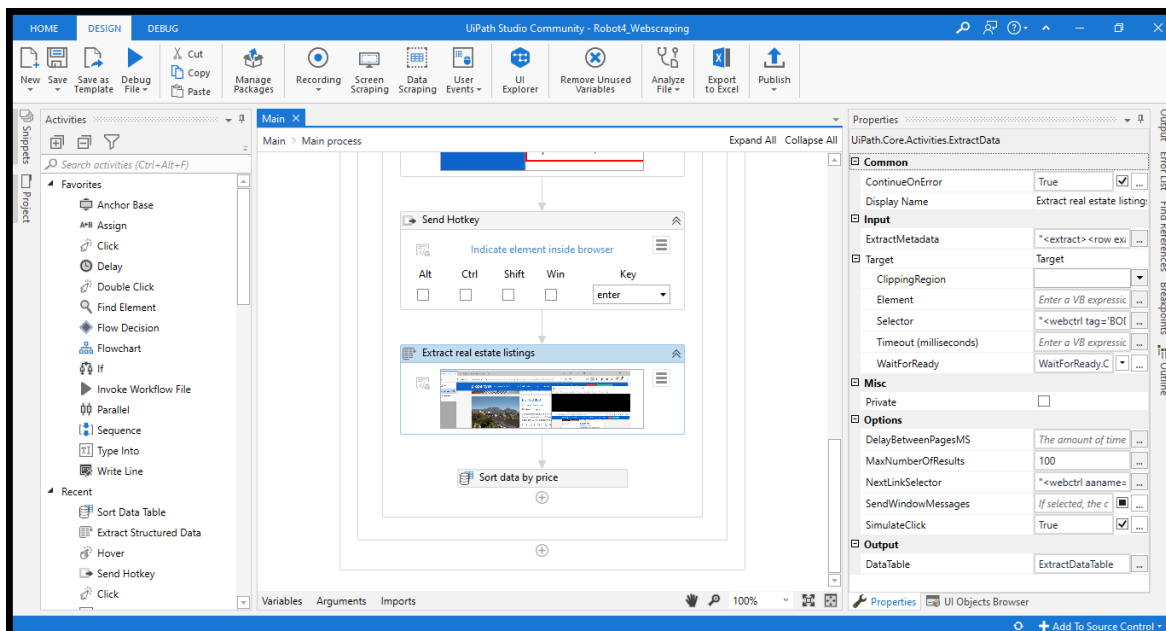


Figure 3 Hands-on Implementation of Document Understanding and Data Validation in UiPath

Reflective Critique:

- Active participation was observed during practical implementation.
- Understanding of scraping concepts improved through hands-on learning.
- Analytical and logical thinking skills were strengthened during workflow creation.

Benefit of the Practice:

- Better understanding of scraping concepts through practical implementation.
- Enhanced ability to identify data extraction requirements.
- Improved workflow design and analytical thinking skills.

- Stronger connection between theoretical concepts and real-world applications.

Challenges faced in implementation:

- Difficulty in identifying suitable web elements during scraping.
- Time constraints during workflow design and implementation.
- Variation in technical understanding among participants.

References:

<https://www.niu.edu/citl/resources/guides/instructional-guide/experiential-learning.shtml>
https://en.wikipedia.org/wiki/Experiential_learning
<https://experientiallearninginstitute.org/what-is-experiential-learning/>

Signature of Faculty Member

HOD